

Pre-Authorization Screening of EIP-7702 Delegates under an Adaptive Adversary

Abstract—EIP-7702 lets an Ethereum externally owned account delegate code execution to a smart contract, creating a security decision at authorization time when behavioral history, reputation, and verified source code may all be unavailable while the delegate’s runtime bytecode is already observable. We study whether that bytecode alone supports pre-authorization risk screening, and we argue that the usual measure of such a screener—clean classification accuracy—does not answer the question that matters. On an audited, family-disjoint benchmark of 2,190 observed delegates in 790 bytecode-similarity families, four very different screeners reach comparable clean performance, including a 15-feature logistic regression that is statistically indistinguishable from our neural model. Under a query-budgeted adaptive adversary restricted to structurally valid, bounded-overhead bytecode edits, they separate sharply: attack success rises to 0.986 for a histogram gradient-boosting model and 0.945 for a flat convolutional encoder, while our hierarchical chunk-attention model, AuthGuard-Seq, holds at 0.181. Crucially, a fixed-transformation study—the standard protocol in prior bytecode security learning—cannot distinguish AuthGuard-Seq from the cheap emulator at all (+0.082, 95% CI [−0.047, +0.218]); only adaptive search resolves them (+0.360, CI [+0.188, +0.518]). Parameter-matched ablations attribute the difference to length-invariant attention aggregation rather than to sequence coverage or hierarchy as such. We further report the first operational measurement of such a screener on live delegation traffic—752 Ethereum delegates observed over a two-week window disjoint from the benchmark at the bytecode-family level—and a measured $652\times$ cost gap against a reference decompiler that motivates millisecond screening ahead of selective deep analysis.

Index Terms—EIP-7702, Ethereum, smart contract security, bytecode analysis, adversarial robustness, trust in delegated execution, security evaluation methodology

I. INTRODUCTION

Ethereum traditionally separates externally owned accounts (EOAs), controlled by user keys, from smart contracts that carry executable logic. EIP-7702 changes this by letting an EOA delegate code execution while keeping its address [1]. This enables transaction batching, gas sponsorship, and programmable authorization, but it also moves execution authority over an account’s assets, approvals, and transaction behavior to third-party code. Authorizing an unsafe delegate is therefore a trust decision made under information scarcity: a newly deployed delegate may have no transaction history, no reputation, and no verified source, yet its runtime bytecode can already be fetched from chain. Figure 1 shows this window.

This paper asks whether bytecode alone supports screening at that boundary, and how such a screener should be evaluated. Our central claim is that the second question determines the answer to the first. Pre-authorization screening is an adver-

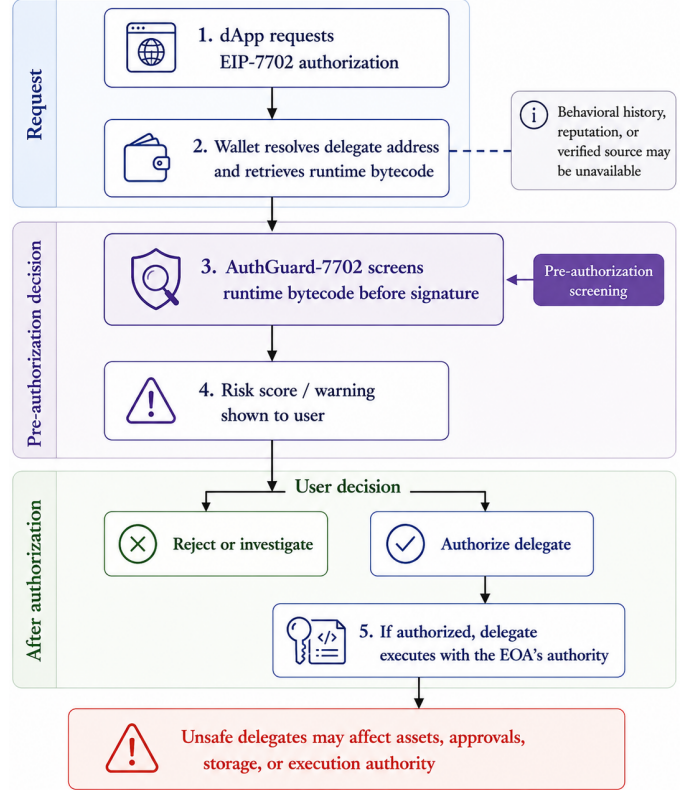


Fig. 1. EIP-7702 authorization flow and the pre-signature screening window.

sarial setting by construction: the attacker writes the bytecode being screened and can pad, reorder, or rewrite it for little more than deployment gas. A screener that is accurate on unmodified code but collapses under cheap edits provides no protection against the adversary it exists to stop.

We find this distinction is not hypothetical. Four screeners spanning gradient boosting, flat convolution, hand-engineered control-flow features, and hierarchical attention reach broadly similar clean performance on our benchmark, and a 15-feature logistic regression is statistically indistinguishable from our neural model ($\Delta\text{AUPRC} -0.021$, 95% CI [−0.078, +0.035]). Under a query-budgeted adaptive attacker they separate by a factor of five. More pointedly, the protocol used throughout prior bytecode security learning—evaluating on one fixed transformation—cannot separate our model from the cheap emulator at all, while adaptive search separates them decisively. The evaluation instrument, not the model, is what resolves the question.

We present AuthGuard-7702, a bytecode-only screening

framework producing a calibrated score and warning tier, built around AuthGuard-Seq: a compact hierarchical model that disassembles runtime bytecode, splits the opcode stream into 256-token chunks, encodes local order with shared convolutions, and aggregates chunk evidence with learned attention. It uses 63,266 active parameters, processes up to 16,384 opcodes, and consumes no transaction history, source code, address identity, or reputation metadata.

A second concern is dependence among contract samples. Code duplication inflates measured performance of learned models of code [7], and contract corpora contain redeployments and near-duplicate variants that let row-level random splits place related programs on both sides of a split [8]. We audit the source data, build bytecode-similarity families, keep identical and related bytecode within one fold, and exclude observations with unreliable source verdicts.

The study makes three contributions.

- **An adversarial evaluation methodology for bytecode security classifiers, and the design finding it yields.** We contribute a query-budgeted adaptive attack over a defined bytecode action space with bounded byte overhead, per-candidate structural validity checking, and a bounded execution-preservation audit, applied across four architecture families with cross-model transfer. It shows that fixed-transformation evaluation is insufficient to rank screeners, and parameter-matched ablations attribute robustness specifically to length-invariant attention aggregation.
- **An audited benchmark and evaluation protocol, with a live temporal holdout.** AuthGuardBench-7702 combines 2,190 audited delegates in 790 families under family-disjoint cross-validation with external benign controls, an audited-implementation registry, and 752 independently collected live Ethereum delegates from a strictly later two-week window, disjoint at the family level. We release the protocol, split and family assignments, per-row predictions, donor provenance, and deterministic reconstruction scripts; redistribution of the third-party bytecode itself is pending license review, so we are careful to claim a reproducible evaluation pipeline rather than a distributable dataset.
- **Operational evidence for staged triage.** We measure the complete local screening path at 4.121 ms median against a pinned reference decompiler at 2.687 s median—a $652\times$ gap—and report flag rates on live delegation traffic weighted by observed authorization volume.

We do not claim architectural novelty. Every component of AuthGuard-Seq is standard; the contribution is the adversarial requirement that motivates the design, the evaluation that isolates which mechanism satisfies it, and the honest negative controls that bound what the model adds.

II. BACKGROUND AND RELATED WORK

A. EIP-7702 and the Authorization Boundary

EIP-7702 introduces authorization tuples through which an EOA delegates code execution while keeping its address.

TABLE I
COMPARISON WITH REPRESENTATIVE PRIOR WORK

Work	EIP-7702	Bytecode ML	Pre-auth.	Adaptive adv.
Huang et al. [2]	✓	–	–	–
Qi et al. [3]	✓	–	–	–
PhishingHook [4]	–	✓	–	–
Eth2Vec [6]	–	✓	–	–
ContractWard [5]	–	✓	–	–
SmartBugBert [9]	–	✓	–	–
AuthGuard-7702	✓	✓	✓	✓

“Adaptive adv.” denotes evaluation against an attacker who searches over input modifications with knowledge of the screener’s scores, as opposed to a fixed predefined transformation.

For each valid tuple a delegation indicator formatted as `0xef0100||address` is written to the authorizing EOA’s code field, and subsequent code-executing operations resolve and run the code at the referenced delegate [1]. The object worth screening is therefore the delegate’s runtime bytecode, not the 23-byte indicator. Once authorized, that code executes in the EOA’s context, which makes the pre-signature moment the natural automation point.

Huang et al. combine transaction analysis with cross-contract static analysis to characterize EIP-7702 malicious behavior [2]; Qi et al. study delegation-based phishing and protocol defenses [3]. Neither investigates learned screening of proposed delegate bytecode before authorization.

B. Learning from Smart-Contract Bytecode

ContractWard uses opcode bigrams with conventional learners [5], Eth2Vec learns contract-wide bytecode representations [6], PhishingHook evaluates bytecode and opcode representations for phishing-contract classification [4], and more recent work combines bytecode features with control-flow-graph analysis [9]. Surveys identify representation, dataset construction, and evaluation design as recurring determinants of reported performance [10].

These works share an evaluation posture: they report clean classification metrics and do not evaluate an adversary who controls the input program. That is the gap this paper addresses. Since the attacker authors the bytecode, clean metrics measure the easy case. Table I summarizes the positioning.

C. Static Analysis and Security-ML Evaluation

Learned screening complements rather than replaces semantic analysis. Gigahorse decompiles EVM bytecode for declarative analysis [11] and Securify derives semantic dependencies and checks security patterns [12]. Our reference labels originate from a decompiler-based rule [2]; our question is whether that signal can be approximated from opcode sequences fast enough for interactive screening, and whether such an approximation survives an adversary.

Methodology matters as much as modeling. TESSERACT shows how dependence and distribution bias inflate security-classification results [8]. We isolate identical and similar bytecode families across folds, and we interpret transformed executable inputs only within their demonstrated preservation guarantees, following problem-space adversarial-ML guidance [13].

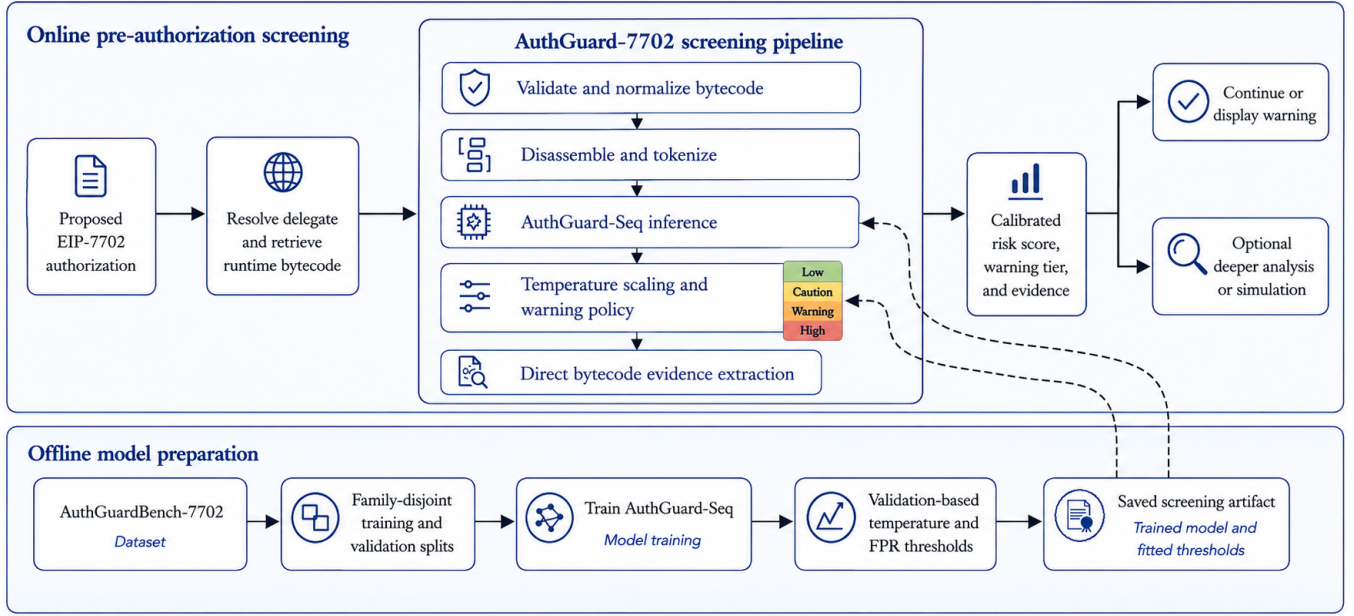


Fig. 2. AuthGuard-7702 system overview: the online screening path at the authorization boundary and the offline family-disjoint training and calibration path.

III. PROBLEM FORMULATION AND THREAT MODEL

A. Pre-Authorization Screening Task

Let b be the normalized runtime bytecode at a proposed delegate address and $T(b)$ its opcode-token sequence. A trained model f_θ produces a risk logit that is temperature-scaled on validation data [14]:

$$s(b) = \sigma\left(\frac{f_\theta(T(b))}{T_{\text{cal}}}\right), \quad T_{\text{cal}} > 0, \quad s(b) \in [0, 1]. \quad (1)$$

The positive class consists of delegates flagged by the source study’s static-analysis rule, which identifies decompiled reachability from standardized inbound hooks—`receive()`, `fallback()`, and token-reception callbacks—to predefined value-moving external-call patterns [2]. The negative class is delegates from the same pool that were not flagged. Because an unflagged verdict does not establish benign semantics, the task is screening for source-analyzer-identified structural risk, not proving malicious intent, exploitability, or safety. We use “source-flagged” rather than “malicious” throughout.

This label provenance is a real limitation, and we address it in two ways rather than one. First, we scope every claim to the reference condition. Second, and more importantly, our headline result is a *relative* comparison among architectures evaluated against a shared label under a shared adversary. Whatever the label denotes, the ranking of screeners under attack is a statement about the screeners, and does not require the label to be adjudicated ground truth. This is the principal reason we organize the paper around adversarial robustness rather than absolute detection quality.

From negative validation scores we derive thresholds $\tau_1 \geq \tau_5 \geq \tau_{10}$ for nominal 1%, 5%, and 10% false-positive-rate (FPR) operating points, mapping scores to HIGH, WARNING,

CAUTION, and LOW-OBSERVED-RISK. Because thresholds are estimated on finite validation sets, realized FPR can differ from nominal, and Section VI-C measures how far.

B. Information Boundary

The model observes only the ordered opcode sequence derived from runtime bytecode. PUSH immediate operands are skipped, so the representation captures opcode identity and order rather than embedded constants. The model receives no transaction history, balances, approvals, graph or address identity, source code, family or fold identifiers, or any field used to construct the reference labels. This boundary is deliberate: the intended case is a delegate with no historical or identity evidence. AuthGuard-7702 is an advisory first-stage screener; it does not resolve proxy state, simulate transactions, infer off-chain intent, or certify safety.

C. Adversary Model

We consider an adversary who wants a delegate whose runtime code exhibits the reference risk condition to be authorized, and who therefore wants that code to score below the wallet’s warning threshold. The adversary **authors the bytecode**. We grant them:

- *Score access.* They may query the screener and observe its output, as they can for any locally deployed or publicly reachable model. We bound this at 64 queries per contract.
- *Edit capability.* They may append content, rewrite metadata, and rewrite embedded address and selector immediates, composing up to four such actions.
- *A cost budget.* Edits are capped at $2\times$ the original executable size, since deployment cost scales with code size.

We do *not* grant white-box gradient access or the ability to retrain the defender’s model. The defender is a wallet or security service that fetches the proposed delegate’s runtime bytecode and applies AuthGuard-7702 before authorization; we assume correct retrieval and trusted model parameters. Compromise of the wallet, signing key, retrieval infrastructure, or screener is out of scope.

Where this model is and is not adequate: Score access with a bounded query budget is the right model for a screening service, where the adversary observes verdicts but not weights. It is deliberately conservative for a screener shipped inside a wallet client, where a determined adversary can extract the parameters and mount white-box gradient attacks. We make this boundary explicit rather than eliding it: the results below are lower bounds on evadability for client-side deployment, and the strongest claim they support is a *relative* one—how architectures compare when the attacker can search over bytecode edits. Closing the white-box case is necessary future work before a client-side deployment claim would be justified, and we do not make one.

The model can only detect risk observable from runtime bytecode. Behavior depending on dynamic state, transaction context, unresolved proxy targets, external counterparties, or off-chain intent may remain invisible.

IV. AUTHGUARD-7702

A. Screening Workflow

Figure 2 places AuthGuard-7702 at the authorization boundary. A wallet resolves the proposed delegate address, retrieves runtime bytecode, and invokes the local path before the user signs. The pipeline normalizes and disassembles the bytecode, runs AuthGuard-Seq, applies validation-fitted calibration and thresholds, and returns a score, warning tier, and auxiliary bytecode evidence. Offline, the training path constructs family-disjoint folds, trains the model, and fits the temperature and FPR thresholds on validation data only; the resulting artifact is what the online path loads.

Auxiliary evidence is extracted separately from raw bytecode and includes code length, opcode counts, call-family instructions, storage writes, contract-creation instructions, and recognized token-transfer or approval selectors. These observations and the learned attention weights indicate influential regions; they are not causal explanations or control-flow proofs.

B. Design Under an Adversarial Requirement

The design follows from the adversary model rather than from architectural ambition. Because the attacker controls program length and can append arbitrary content cheaply, the aggregation step must not let appended bytes dilute or dominate the risk signal. This yields three requirements, each tied to an attack and each tested in Section VI-A:

- 1) *Full-stream coverage.* Fixed-window models can be evaded by pushing the payload past the window, so the model must see the whole program rather than a prefix.

- 2) *Local order sensitivity.* Histogram and n -gram representations discard instruction order, which order-preserving rewrites exploit; local convolutions retain short ordered motifs.
- 3) *Length-invariant, selective aggregation.* Mean pooling over a longer program dilutes evidence proportionally to appended content. Learned attention can concentrate on the informative region regardless of how much padding surrounds it.

C. Hierarchical Opcode-Sequence Model

AuthGuard-Seq begins with deterministic linear-sweep EVM disassembly. Recognized opcodes map to stable token IDs, undefined bytes use a dedicated token, and padding uses a separate ID. PUSH immediates are skipped by instruction width. Tokenization does not terminate at the first linear STOP, so trailing regions are not discarded because of an early stopping opcode. The sequence is split into chunks of length $L = 256$, at most $K = 64$ chunks. The longest benchmark program has 16,081 opcodes and fits the 16,384-token capacity, so every benchmark contract is processed in full; longer contracts outside the benchmark retain evenly spaced chunks rather than a prefix.

Each chunk is embedded into 32 dimensions, processed by a kernel-5 convolution followed by a kernel-3 dilated convolution with dilation 2, and reduced by masked maximum pooling to a 64-dimensional h_k . Learned attention aggregates valid chunks:

$$\alpha_k = \frac{\exp(w^\top h_k)}{\sum_{j=1}^{K_b} \exp(w^\top h_j)}, \quad h = \sum_{k=1}^{K_b} \alpha_k h_k, \quad (2)$$

where w is learned and K_b is the number of non-padding chunks. The aggregate passes through a 128-dimensional layer with GELU, dropout, layer normalization, and a binary risk head. The logit is temperature-scaled before thresholds apply. Figure 3 summarizes the architecture.

Parameter accounting: AuthGuard-Seq is the sequence-only configuration of a fusion architecture that also defines n -gram and dense-structural branches. Those branches are constructed but receive no gradient in this configuration, so a naive parameter sum over the module overstates the model and, worse, reports an identical count for configurations that differ. We report parameters that are actually on the trained forward path, verified by checking which tensors receive gradient: **63,266** for AuthGuard-Seq, against 108,225 for the n -gram variant and 75,595 for the dense-structural variant. The serialized checkpoint used for timing is 742,625 bytes because it stores all constructed branches.

V. BENCHMARK AND METHODOLOGY

A. Audited Benchmark Construction

AuthGuardBench-7702 contains 3,082 audited observations partitioned before evaluation (Table II). The primary population derives from the EIP-7702 delegate dataset and analyzer outputs of Huang et al. [2]; after auditing it holds 2,190

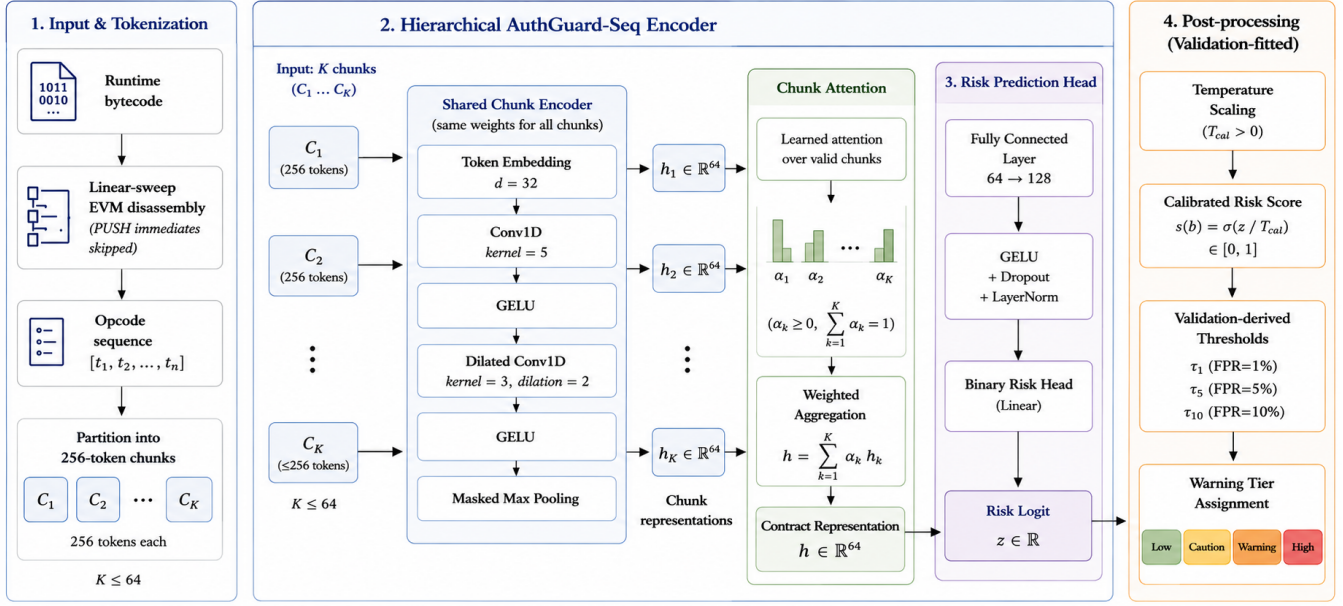


Fig. 3. AuthGuard-Seq: chunked opcode encoding, shared local convolutional encoders, and learned chunk attention. Attention aggregation is the mechanism the ablation in Table IV isolates as responsible for robustness under bytecode flooding.

TABLE II
AUDITED BENCHMARK POPULATIONS

Population	n	Reference status	Families
Primary evaluation	2,190	727 flagged / 1,463 unflagged	790
External benign control	797	Benign-labeled	—
Audited implementations	8	Named production delegates	—
Excluded uncertain input	90	Unreliable source verdict	—
Live temporal holdout	752	Unlabeled	454

delegates, 727 source-flagged and 1,463 unflagged. External controls contribute 797 benign-labeled general Ethereum contracts from the PhishingHook-derived source [4], and eight audited production EIP-7702 implementations provide named qualitative controls. A further 90 observations are excluded because their original unflagged verdicts came from truncated or failed bytecode retrieval: we identified 89 negatives truncated at a spreadsheet cell limit and one timeout string stored in place of bytecode. Although the runtimes were recoverable, the verdicts had not been established on repaired inputs.

Section VI-C adds a fourth population collected independently for this work: 752 screenable Ethereum delegates observed in a two-week window strictly later than the benchmark.

B. Families, Duplicates, and Splits

Runtime bytecode is linearly disassembled and represented as opcode 4-gram sets. Deterministic 128-permutation BLAKE2b MinHash signatures with a fixed 0.85 agreement threshold define similarity links, merged by union-find into 790 families. Identical bytecode with conflicting reference labels is quarantined before modeling.

We use five fixed family-disjoint outer folds. For test fold f , fold $(f + 1) \bmod 5$ is validation and the remaining three

are training. No family or identical bytecode crosses folds. Each configuration is repeated with seeds 7702, 7703, and 7704; results average five folds within a seed and then report mean and sample standard deviation across the three seed-level means.

C. Adaptive Attack Protocol

The attack searches over compositions of eight actions: metadata rewrite; embedded address rewrite; function-selector rewrite; neutral instruction insertion; and flooding that appends a STOP followed by deterministic donor bytes at 25%, 50%, 100%, or 200% of the source executable length. At most one flooding action and at most four actions total may compose.

Two search strategies share a 64-query budget per contract: a seeded *random search* over sampled action sequences, and a score-guided *beam search* with width 4 and depth 4 that expands the lowest-scoring candidates. Both are deterministic given the seed. We additionally evaluate each fixed transformation alone and a per-source oracle over the seven fixed conditions.

Three constraints keep the attack honest. Every candidate must be **structurally valid**, i.e. disassemble and satisfy an opcode-skeleton preservation check against the source. Every candidate must respect a $2\times$ **byte-overhead cap**. Flooding donors are drawn from partition-isolated pools such that a donor never shares the recipient’s bytecode family and never crosses the train/validation/test boundary, with full provenance recorded per segment.

An attack *succeeds* on a source if the clean score is at or above the model’s own validation-derived 5% FPR threshold and the adversarial score falls below it. Attack success rate (ASR) is computed over sources the model detects cleanly,

since a source it never flagged cannot be evaded. Each architecture is attacked as the target in turn, and each target’s best candidate is scored by the other models to give transfer rates.

Execution preservation: A bounded audit of 100 calls across ten delegates compares source and transformed execution. Flooding preserved recorded execution behavior in 100/100 calls. Address-immediate rewriting preserved only 23/100, so conditions involving it are reported as representation stress, not behavior-preserving, and we mark them accordingly. We claim bounded empirical preservation for flooding, not universal semantic equivalence.

D. Baselines, Training, and Statistics

The clean comparison evaluates seven predefined configurations under identical folds, validation rotation, seeds, early stopping, calibration, and operating points. Nonhierarchical neural baselines use a 2,048-position opcode view obtained by uniform sampling across the whole stream; AuthGuard-Seq processes complete streams in at most 64 chunks of 256 tokens. Neural models use class-weighted binary cross-entropy with positive-class weight equal to the training-fold negative-to-positive ratio, AdamW at learning rate 10^{-3} , weight decay 10^{-4} , gradient clipping at 5, at most 30 epochs, and validation-AUPRC early stopping with patience 5. Temperature scaling is fit on validation logits only and thresholds derive only from negative validation examples. Test labels never influence model selection, calibration, or thresholds.

AUPRC is the primary ranking metric because the population is imbalanced and the metric emphasizes positive-class ranking; AUROC and calibrated Brier score are secondary. Operational behavior is Recall@5% FPR. For inferential comparisons the bytecode family is the resampling unit: families are resampled with replacement within each outer test fold and the same multiplicities apply to both members of a pair. **Replicate counts differ by experiment and we state them per table:** 10,000 for primary clean comparisons, 2,000 for the mechanism ablation, and 1,998 for the cheap-baseline controls.

VI. EVALUATION

We ask four questions. RQ1: what does clean screening actually require, and which mechanism supplies it? RQ2: which screeners survive an adversary who writes the bytecode? RQ3: do calibrated operating points transfer across time and population? RQ4: is staged triage operationally justified?

A. RQ1: Clean Screening and What It Fails to Distinguish

Table III reports clean family-disjoint results. AuthGuard-Seq ranks first at 0.924 ± 0.014 AUPRC and 0.833 ± 0.016 Recall@5% FPR with held-out FPR 0.052 ± 0.007 . Stated operationally, that recall means roughly **one in six source-flagged delegates is not surfaced** at the recommended operating point even on unmodified bytecode—a first-stage triage layer, not a gate. Family-clustered paired bootstrap over 10,000 replicates supports the margin over Flat CNN ($+0.039$ AUPRC, CI $[+0.009, +0.073]$; $+0.121$ recall, CI $[+0.050, +0.190]$)

TABLE III
CLEAN FAMILY-DISJOINT PERFORMANCE. PARAMETERS FOR FUSION CONFIGURATIONS ARE ACTIVE-PATH COUNTS (SECTION IV).

Model	Params	AUPRC $\uparrow$	AUROC $\uparrow$	Brier $\downarrow$	R@5% $\uparrow$
AuthGuard-Seq	63,266	.924$\pm$.014	.963$\pm$.011	.072$\pm$.012	.833$\pm$.016
Flat CNN	154,177	.885 $\pm$.010	.937 $\pm$.007	.099 $\pm$.004	.712 $\pm$.024
Hist.+4-gram XGB	—	.833 $\pm$.004	.907 $\pm$.002	.127 $\pm$.004	.615 $\pm$.015
Neural n -gram	108,225	.810 $\pm$.007	.880 $\pm$.020	.125 $\pm$.001	.654 $\pm$.029
BiGRU	108,225	.679 $\pm$.098	.815 $\pm$.069	.163 $\pm$.027	.379 $\pm$.113
Dense structural	75,595	.637 $\pm$.018	.780 $\pm$.022	.182 $\pm$.009	.331 $\pm$.023
Transformer	425,217	.563 $\pm$.031	.730 $\pm$.019	.208 $\pm$.013	.239 $\pm$.054

and over histogram+ n -gram XGBoost ($+0.091$ AUPRC, CI $[+0.045, +0.140]$).

A cheap baseline matches the neural model: We deliberately construct a strong trivial competitor: an L2 logistic regression over 15 hand-coded opcode and control-flow features, including a conservative and an over-approximate variant of “does a fallback path reach an external call.” It reaches 0.911 AUPRC against AuthGuard-Seq’s 0.924, and the paired family-clustered difference is -0.021 with 95% CI $[-0.078, +0.035]$, which **spans zero**. It also runs faster: 2.626 ms versus 4.121 ms median. A single boolean—over-approximate fallback-to-external-call reachability—achieves recall 0.996 at precision 0.760 on its own.

We report this as a result, not a caveat. On clean bytecode the reference condition is cheaply recoverable, and clean accuracy alone does not justify a learned sequence model. A memorization control confirms the split is sound rather than the task being trivially near-duplicate: k -NN over Min-Hash signatures reaches only 0.612 AUPRC, -0.302 against AuthGuard-Seq with CI $[-0.380, -0.226]$.

Which mechanism matters: Table IV reports a separate controlled study in which five variants are matched at roughly 30,000 parameters, isolating token budget, layout, and aggregation, and evaluated both clean and under Flood-200%. Two findings stand out. On clean bytecode, a flat encoder given the same 16,384-token budget is at least as good as chunk attention—the hierarchy contrast is -0.018 with CI $[-0.040, +0.018]$, spanning zero—so we do not claim clean superiority for hierarchy. Under flooding the ordering inverts sharply: attention over mean pooling gives $+0.180$ AUPRC (CI $[+0.135, +0.232]$) and $+0.541$ Recall@5% (CI $[+0.474, +0.631]$), and chunking over flat gives $+0.098$ (CI $[+0.059, +0.154]$). Coverage alone is inconclusive in both conditions.

The clean column of Table IV is the paper’s thesis in miniature: the best clean model is the worst model under attack.

The mechanism, measured directly: The ablation establishes *that* attention aggregation matters; it does not show *why*. The design argument in Section IV is that attention can concentrate on the informative region while uniform weighting must dilute evidence in proportion to appended content. Because the model exposes its chunk weights, we can test this rather than infer it. For all 727 held-out positives, scored by each fold’s own checkpoint, we recorded the fraction of attention mass falling on chunks composed entirely of

TABLE IV

PARAMETER-MATCHED MECHANISM ABLATION (SEPARATE CONTROLLED RUN; 3 SEEDS $\times$ 5 FOLDS; CONTRASTS USE 2,000 FAMILY-CLUSTERED REPLICATES).

Variant	Params	Clean	F200	F200 R@5%
Flat control (2K)	29,985	.879 $\pm$.005	.532 $\pm$.017	.148 $\pm$.013
Flat control (16K)	29,985	.936$\pm$.003	.810 $\pm$.014	.506 $\pm$.038
Chunk attention (2K)	30,050	.902 $\pm$.003	.897 $\pm$.009	.817 $\pm$.022
Chunk mean (16K)	29,985	.879 $\pm$.005	.728 $\pm$.020	.274 $\pm$.031
Chunk attention (16K)	30,050	.918 $\pm$.007	.908$\pm$.001	.815$\pm$.039
AuthGuard-Seq (deployed)	63,266	.914 $\pm$.013	.894 $\pm$.006	.786 $\pm$.021
<i>Mechanism contrasts (ΔAUPRC, 95% CI)</i>				
Coverage (16K–2K)		Clean +.015 [−.003, +.030]	Flood-200% +.011 [−.005, +.025]	
Attention (−mean)		+.039 [+.007, +.064]	+.180 [+.135, +.232]	
Hierarchy (−flat)		−.018 [−.040, +.018]	+.098 [+.059, +.154]	

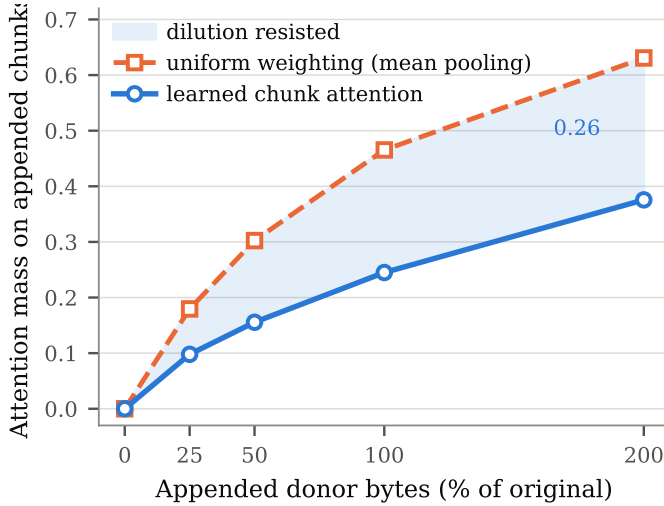


Fig. 4. Attention mass landing on appended donor chunks, against the uniform-weighting baseline that mean pooling is forced to. All 727 held-out positives, each scored by its own fold’s checkpoint. The shaded gap is the dilution the learned aggregation resists.

appended donor bytes. Uniform weighting has no learned parameters, so its mass on appended chunks is exactly their share of the chunk count—the analytic baseline the chunk-mean control realizes empirically.

Figure 4 shows the result. Under Flood-200%, appended content occupies 63.1% of chunks, but attention places only 37.6% of its mass there, resisting 25.5 percentage points of dilution; the gap widens monotonically with flooding volume (8.2, 14.7, 22.1, and 25.5 points at 25%, 50%, 100%, and 200%). The mechanism is real and it is also *partial*: attention still assigns more than a third of its weight to content that carries no signal, which is a direct explanation of why the residual attack success rate in Section VI-B is 0.181 rather than zero.

B. RQ2: Robustness to an Adaptive Adversary

We attacked four architectures across three seeds and five folds under the protocol of Section V-C, producing 87,240

TABLE V

ATTACK SUCCESS RATE BY TARGET AND STRATEGY (727 SOURCES, 64-QUERY BUDGET, $\leq 2\times$ BYTE OVERHEAD, 100% STRUCTURAL VALIDITY). LOWER IS BETTER. POOLED ASR WITH FAMILY-CLUSTERED 95% CI. UPPER BLOCK: 3 SEEDS $\times$ 5 FOLDS. MIDDLE AND LOWER BLOCKS: SEED 7702, 5 FOLDS.

Target	Clean det.	Flood-200%	Beam	Random
AuthGuard-Seq	.843	.120	.146	.181
15-feature emulator	.825	.182	.452	.536
Flat CNN	.722	.750	.907	.945
Hist.+4-gram XGBoost	.637	.677	.950	.986
<i>With flooding-augmented training</i>				
AuthGuard-Seq + aug.	.839	.039	.051	.062
Flat CNN + aug.	.644	.682	.882	.927
Hist.+4-gram XGB + aug.	.670	.530	.840	.940
<i>Parameter-matched controls ($\approx 30K$ params, cf. Table IV)</i>				
Chunk attention 16K	.847	.045	.068	.075
Flat control 16K	.831	.422	.705	.889
Chunk mean 16K	.769	.723	.923	.986

attack records over 727 source contracts. Table V reports attack success.

AuthGuard-Seq is simultaneously the best clean detector (0.843 detected at the 5% operating point) and by far the hardest to evade (0.181 under the strongest attack). Paired family-clustered contrasts against AuthGuard-Seq under random search are +0.786 (CI [+0.727, +0.838]) for Flat CNN and +0.844 (CI [+0.786, +0.890]) for XGBoost.

Random search beats guided search: On every target, seeded random search outperforms score-guided beam search (e.g. 0.181 versus 0.146 against AuthGuard-Seq, 0.986 versus 0.950 against XGBoost). The beam follows first-step score reductions into a local optimum while random search explores more diverse compositions on the same budget. We report the *maximum* over strategies as the attack result, so a weak guided search cannot flatter the defence: the reported ASR is the best of two budget-matched attackers, and it is the undirected one that is stronger here. This also means the numbers in Table V should be read as lower bounds on what a better-designed search would achieve.

Fixed transformations cannot rank these models: The decisive comparison is between AuthGuard-Seq and the cheap emulator, which were statistically tied on clean data. Table VI shows their gap as a function of attack strength.

Under a single fixed transformation—the protocol used throughout prior bytecode security learning, and in the earlier version of this work—a 15-feature logistic regression and a hierarchical neural model are **indistinguishable**. The separation appears only under adaptive search and grows monotonically with attack strength, as Figure 5 shows. This is the strongest

TABLE VI
EMULATOR — AUTHGUARD-SEQ ASR, BY ATTACK STRENGTH. PAIRED,
FAMILY-CLUSTERED, 1,512 ELIGIBLE SOURCES.

Attack strength	Δ ASR (95% CI)	Excludes 0
Fixed Flood-200%	+0.082 [-0.047, +0.218]	no
Best of 7 fixed transforms	+0.221 [+0.067, +0.369]	yes
Beam search, 64 queries	+0.330 [+0.161, +0.488]	yes
Random search, 64 queries	+0.360 [+0.188, +0.518]	yes

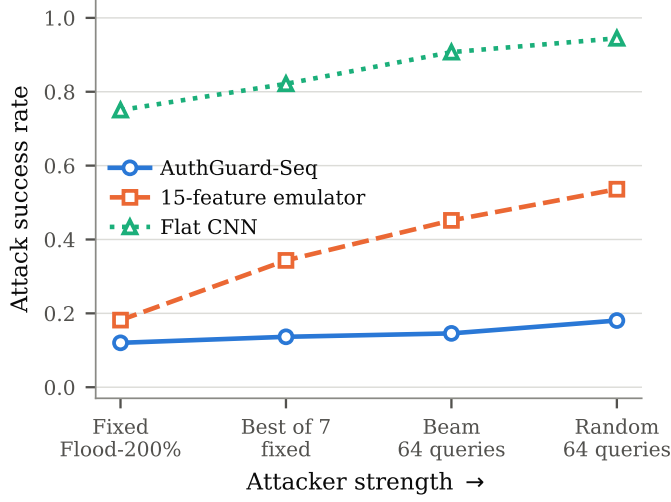


Fig. 5. Attack success against attacker strength. The models that are close under a fixed transformation separate under budgeted adaptive search.

justification we can offer for the methodology itself: it is not decoration around a model result, it is the only instrument in this study that resolves the comparison.

Can training substitute for architecture?: The natural objection to an architectural claim is that robustness might come from training data rather than design: augment the baselines with flooded examples and the gap may close. We tested this directly. Each model was retrained with its training set doubled by one flooding-augmented copy of every training contract, drawn from train-partition donors with fractions cycled over 25–200% so the augmented set spans the flooding range. Results are the lower block of Table V.

Augmentation does help—against the transformation it was trained on. Under fixed Flood-200% it cuts Flat CNN from 0.750 to 0.682 and XGBoost from 0.677 to 0.530. Under adaptive search it barely moves: Flat CNN 0.945 to 0.927, XGBoost 0.986 to 0.940. Both augmented baselines remain worse than clean-trained AuthGuard-Seq by +0.774 (CI [+0.697, +0.841]) and +0.786 (CI [+0.703, +0.852]) respectively, intervals excluding zero. Augmentation therefore does not substitute for the architecture. It is also not free for the baselines: Flat CNN loses clean detection (0.722 to 0.644) while buying almost no adaptive robustness—the familiar robustness–accuracy tension.

The same intervention behaves completely differently on the attention model. AuthGuard-Seq falls from 0.181 to **0.062** under adaptive search (−0.133, CI [−0.194, −0.086]) at essentially no clean cost (0.843 to 0.839). Augmentation and atten-

TABLE VII
CROSS-MODEL TRANSFER OF BEAM-SEARCH ATTACKS. ROW: MODEL THE
ATTACK WAS OPTIMIZED AGAINST. COLUMN: MODEL SCORING THE
RESULTING BYTECODE.

Optimized against ↓ / scored by →	AuthGuard-Seq	Flat CNN	XGBoost
AuthGuard-Seq	—	.782	.673
Flat CNN	.088	—	.596
Hist.+4-gram XGBoost	.084	.652	—

tion aggregation are complementary rather than alternative: the augmented data helps when the architecture can exploit it. We note this is also a third instance of the paper’s methodological point—judged only on fixed Flood-200%, augmentation would look like a substantial defence for the baselines, and under adaptive evaluation it is not.

The mechanism claim holds under adaptive attack:

Section VI-A attributed robustness to attention aggregation using clean and fixed-flooding evaluation. Since we argue fixed transformations are an insufficient instrument, we re-tested that attribution with the adaptive attack on the same parameter-matched controls (lower block of Table V). At roughly 30,000 parameters each, chunk attention reaches 0.075 adaptive ASR against 0.889 for the flat control and 0.986 for chunk mean—a thirteenfold gap to mean pooling at matched capacity, with paired contrasts of +0.837 (CI [+0.775, +0.884]) and +0.710 (CI [+0.636, +0.772]) against AuthGuard-Seq. The mechanism attribution is therefore not an artefact of the weaker protocol.

One result runs against expectation and we report it as such: the 30,050-parameter matched attention control is *more* robust than the 63,266-parameter deployed model (−0.129, CI [−0.189, −0.081]) at equal clean detection (0.847 versus 0.843). On this evidence the smaller matched configuration, not the larger reference architecture, is the one a deployment should use.

Transfer is asymmetric: Table VII scores each target’s best beam-search candidate against the other models. Attacks constructed against AuthGuard-Seq transfer to Flat CNN at 0.782 and XGBoost at 0.673, while attacks constructed against those models transfer to AuthGuard-Seq at only 0.088 and 0.084. Evasive directions found against flat aggregation do not generalize to attention aggregation, though the converse holds freely. This is independent evidence that the robustness gap reflects the representation rather than a fortunate threshold.

For completeness, under the fixed conditions of the earlier protocol AuthGuard-Seq retains 0.920 ± 0.007 AUPRC under Flood-200% and 0.912 ± 0.005 under the combined rewrite-and-flood condition, against 0.535 ± 0.013 and 0.525 ± 0.011 for Flat CNN. As Table VI shows, those numbers are real but not discriminative.

C. RQ3: Transfer Across Populations and Time

External benign controls: Applying primary validation thresholds to 797 benign-labeled general Ethereum contracts yields FPR 0.015 ± 0.004 , 0.065 ± 0.012 , and 0.169 ± 0.021 at the nominal 1%, 5%, and 10% points. The 5% threshold

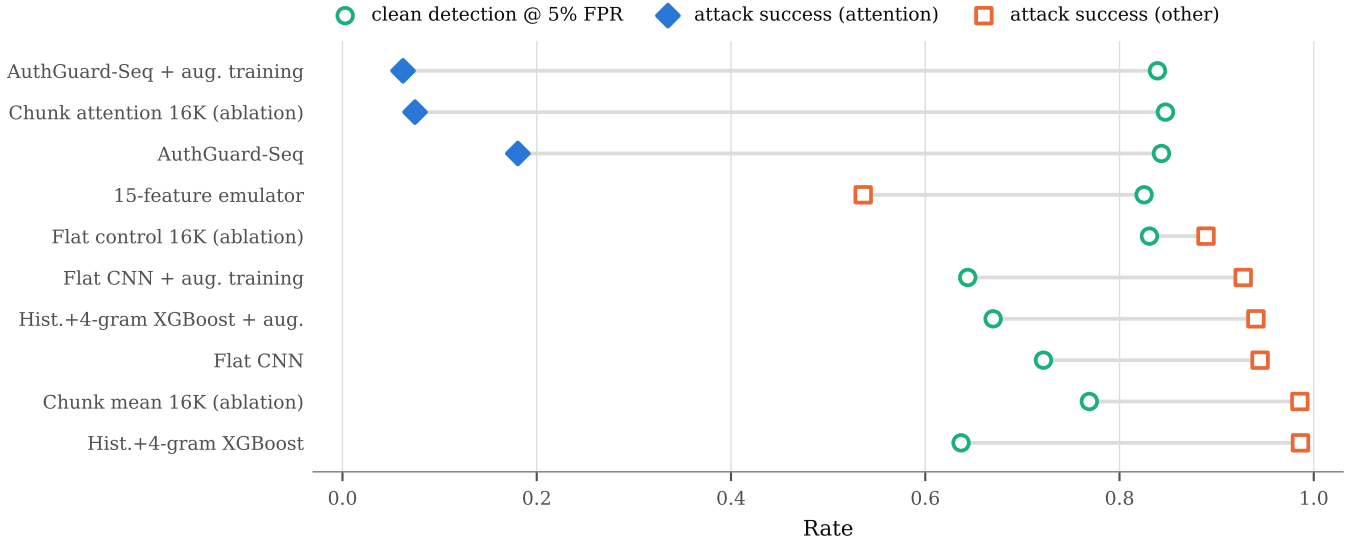


Fig. 6. Clean detection and worst-case attack success for every evaluated configuration, ordered by robustness. Clean detection spans a narrow band and does not order the models; attack success spans almost the full range. The three configurations with attention aggregation occupy the robust end regardless of parameter count or training scheme.

TABLE VIII
LIVE TEMPORAL HOLDOUT: FLAG RATES OVER 752 DELEGATES AND 15 FROZEN CHECKPOINTS. NO LABELS ASSERTED.

Nominal point	By contract	Leakage-clean	Authorization-weighted
1%	9.5%±2.1	8.8%	5.9%
5%	20.3%±3.6	19.9%	9.4%
10%	32.7%±1.9	33.3%	14.6%

transfers with moderate drift; the 10% point degrades substantially.

A live temporal holdout: The benchmark carries no timestamp, so temporal generalization was previously untestable. We therefore collected an independent population: scanning 100,444 Ethereum blocks yielded 1,335,391 authorization entries and 752 screenable distinct delegates first observed between 2026-07-23 and 2026-08-06, strictly later than the benchmark, with 669 unique runtime bytecodes.

Exact bytecode-hash overlap with the entire benchmark is zero. That is not sufficient evidence of novelty: applying the benchmark’s own 0.85 MinHash family threshold, **70 of 752 (9.3%)** live delegates would join a training family, one at similarity 1.000—identical opcode 4-gram profile differing only in immediates. We report the family-level gate and give leakage-clean figures alongside.

We assert no labels for this population. What we measure is operating-point behavior: what fraction of a real delegate stream each frozen checkpoint flags at its own thresholds (Table VIII).

The deployment-relevant figure is the authorization-weighted one, because a wallet’s users encounter delegates in proportion to how often they are actually authorized, not in proportion to how many distinct delegates exist. On that basis the nominal 5% point flags **9.4%** of live authorization volume and the nominal 1% point flags **5.9%**; the latter is a plausible

deployment setting for an advisory tier. The weighted rates are roughly half the per-contract rates because high-volume delegates are predominantly popular wallet-infrastructure implementations that score low, while the long tail of rarely-authorized delegates carries most of the flags.

Two further readings are required. First, thresholds drift substantially: the nominal 5% point flags 20.3% of distinct live delegates, far above the 6.5% seen on curated benign controls, so population-specific recalibration is mandatory before deployment. Second, this stream is real mixed traffic containing genuinely risky delegates, so 20.3% is a flag rate, not a false-positive rate, and cannot be decomposed without labels—we deliberately do not report it as a false-alarm measurement.

Audited implementations: Across eight named production EIP-7702 implementations, three occur within primary training families; scoring current checkpoints on those would leak, so we exclude them from any aggregate and report them only descriptively. Of the remainder, none is flagged at the 5% point, though several fall in the CAUTION band at the 10% point. We treat this as a qualitative check, not a false-alarm estimate; $n = 8$ cannot support one.

D. RQ4: Operational Cost and Staged Triage

Across 1,500 complete screening calls on one CPU thread, the full local path has median latency 4.121 ms, p95 14.547 ms, and p99 21.429 ms; forward inference alone is 0.950 ms median and model loading 7.690 ms, amortizable in a persistent process. Measurements exclude RPC access, node latency, wallet rendering, and user interaction.

The timed artifact is the seed-7702, fold-0 cross-validation checkpoint, used here for cost measurement rather than as a released model. A deployed screener would be a single model retrained on all folds with calibration and thresholds fit on a held-out split drawn from the intended deployment population,

TABLE IX
MEASURED COST OF THE TWO STAGES OF A STAGED PIPELINE.
SCREENING: SINGLE CPU THREAD, AMD RYZEN 5 3600, PyTorch 2.9.
REFERENCE DECOMPILER: PINNED GIGAHORSE IMAGE, 60
FAMILY-DISTINCT INPUTS, ONE JOB.

Scope	Calls	Median	p95
Full local screening path	1,500	4.121 ms	14.547 ms
forward inference only	195	0.950 ms	1.585 ms
model load (amortizable)	10	7.690 ms	9.716 ms
Reference decompiler (warm)	60	2.687 s	5.618 s
cold, incl. Datalog compile	1	264.23 s	–
Median ratio		652×	

following the recalibration finding of Section VI-C; we report cross-validation artifacts throughout because every accuracy and robustness number in this paper is a held-out estimate, and a fully retrained model would have no untouched data left to evaluate it on.

For the deeper-analysis side of a staged pipeline, we measured a pinned official Gigahorse decompiler image on 60 deterministic family-distinct inputs. All 60 succeeded, with warm median 2.687 s, p95 5.618 s, maximum 28.067 s, 210.59 s total warm serial wall time, and 275.7 MB peak memory; a separate cold run including Datalog compilation took 264.23 s at 10.9 GB peak.

The median ratio against the full local screening path is **652×** (Table IX). This supports staged triage: millisecond screening at the authorization boundary, with deeper semantic analysis reserved for escalated cases. Three scoping statements are essential. This measures decompilation and lifting only; **no downstream client rule was executed**. It is not the exact analyzer of Huang et al., and it is not a comparison of accuracy, utility, or substitutability. And the speedup accrues to fast bytecode-level screening in general—the 2.626 ms emulator of Section VI-A is also an instance—so it is not a property unique to AuthGuard-Seq.

VII. DISCUSSION AND LIMITATIONS

A. What the Results Support

The results support a division of labor: rapid learned screening prioritizes delegates for static analysis, symbolic execution, history checks, or human review, rather than acting as a proof engine. They also support a narrower and more useful claim about what such a screener needs. Clean accuracy is achievable by very cheap models, so it is a poor basis for choosing one. Robustness to an adversary who authors the bytecode is not cheaply achievable, separates architectures by a factor of five, and is invisible to the fixed-transformation protocol prior work uses.

Trust establishment under delegated execution: EIP-7702 changes what a signature means. Before it, authorizing a transaction committed a user to one state transition they could in principle inspect; after it, authorizing a delegation commits them to whatever a third-party program does with their account until revoked. That shifts the security question from transaction validation to *counterparty assessment*, and it does so at a moment when the usual bases for trust—history,

reputation, audit status, verified source—are frequently absent by construction, because the delegate may have been deployed minutes earlier. Bytecode screening is best understood as a substitute trust signal for exactly that gap: it cannot establish that a delegate is safe, but it can supply evidence where no other evidence exists yet, and it can do so in milliseconds at the moment the decision is made. This framing also explains why adversarial robustness is the property that matters. A trust signal an adversary can cheaply manufacture is worse than no signal, because it converts absence of warning into false assurance. The same argument applies to any learned trust signal over attacker-authored artifacts, which is why we regard the evaluation methodology rather than the specific model as the transferable contribution.

We emphasize what we do not claim. We claim no architectural novelty; every component is standard. We do not claim clean superiority for hierarchy over a parameter-matched flat encoder, which our own ablation does not support. We do not claim robustness against white-box gradient attacks, adaptive retraining, or arbitrary semantics-preserving transformations, none of which we evaluated.

B. Limitations

Label provenance. Reference labels encode a decompiler-derived structural condition, not adjudicated malicious intent. Our relative, adversarial framing reduces but does not eliminate dependence on that condition: the ranking under attack is a statement about screeners, but the clean numbers and the detected-source pool still inherit the rule’s definition. Independent adjudication by multiple human reviewers remains the right next step and was out of scope here.

Family threshold. The 0.85 MinHash threshold is a choice. A memorization control shows only 1.2% of test rows have a > 0.9 -similar training neighbor, but 43.6% fall in the 0.7–0.9 band, where k -NN reaches 0.66–0.75 AUPRC. Substantial near-duplicate structure therefore crosses folds below the threshold, and reported generalization should be read with that in mind.

Attack scope. The adversary is score-access, bounded-overhead, and restricted to eight actions composed at most four deep with a 64-query budget. Stronger adversaries—gradient-based, higher budget, or with novel action primitives—are unevaluated, and the 0.181 residual ASR means roughly one in five detected risky delegates is already evadable within our own threat model.

Preservation. Flooding preserved execution in 100/100 audited calls, but the audit covers ten delegates. Address-immediate rewriting preserved only 23/100 and is reported as representation stress throughout.

Live population. The temporal holdout is unlabeled, Ethereum-only, and two weeks long, whereas the benchmark spans seven chains. It measures operating-point transfer, not detection quality, and its 9.3% family-level overlap with training bytecode is disclosed rather than removed from the primary figure.

Variance. Three training seeds inform the primary clean and adaptive-attack results. The augmented-training and parameter-matched control blocks of Table V are single-seed (7702) across all five folds, so their confidence intervals reflect family resampling but not seed variability; the effects there are large relative to the seed spread observed elsewhere (at most 0.020 in fold-then-seed ASR), but we do not claim seed-level stability for them. Elsewhere in this project, cuDNN algorithm selection moved a comparable recall metric by up to 0.026, so reported deviations may understate cross-hardware variability.

C. Ethics and Responsible Disclosure

This work analyzes publicly deployed on-chain bytecode and public transaction data; no private data, user accounts, or credentials were accessed, and no contracts were deployed, invoked, or interacted with on any live network. The adaptive attack operates entirely offline on locally stored bytecode against locally trained models. We report attack success rates against our own screeners to establish an evaluation methodology, and the transformation primitives used—appending dead code, rewriting metadata—are already documented in the prior literature we cite and are not novel offensive capability. Reference labels derive from a published academic artifact [2]; we identify no previously undisclosed vulnerable contract, and we name production implementations only as legitimate controls, never as risky. The authors declare no conflicts of interest.

D. Artifact Availability

The evaluation protocol, family and fold assignments, per-row predictions for every experiment, donor provenance ledgers, attack traces, and SHA-256 manifests are prepared for release, together with deterministic reconstruction scripts for the corpus. We do not redistribute the third-party bytecode corpus itself pending license review of the source artifact, and provide reconstruction instructions with content hashes instead. Reproducibility caveats we can state now: three training seeds inform all reported deviations; bootstrap replicate counts differ per experiment as noted per table; and of the four SHA-256 records accompanying the reference-decompiler measurement, the three scientific outputs verify exactly while the manifest’s self-recorded hash is unreproducible because the manifest enumerates the verification file that in turn records it.

VIII. CONCLUSION

We presented AuthGuard-7702, a bytecode-only pre-authorization screening framework for EIP-7702 delegation, and evaluated it against an adversary who writes the bytecode being screened. On an audited family-disjoint benchmark, four architectures reach comparable clean performance and a 15-feature logistic regression is statistically indistinguishable from our neural model. Under a query-budgeted adaptive attacker they separate decisively: attack success reaches 0.986 and 0.945 for gradient-boosting and flat convolutional screeners while AuthGuard-Seq holds at 0.181, with parameter-matched

ablations attributing the difference to length-invariant attention aggregation. The comparison that matters is invisible to fixed-transformation evaluation, which is what prior work in this area reports. We additionally contribute a live temporal holdout of 752 independently collected delegates with family-level disjointness accounting, and a measured $652\times$ cost gap against a reference decompiler that motivates millisecond screening ahead of selective deep analysis. We read these results as evidence that adversarial evaluation should be the default for learned smart-contract security screening, not an optional extension of it.

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